

TokenFlow: Unified Image Tokenizer for Multimodal Understanding and Generation

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Abstract

We present **TokenFlow**, a novel unified image tokenizer that bridges the long-standing gap between multimodal understanding and generation. Prior research attempt to employ a single reconstruction-targeted Vector Quantization (VQ) encoder for unifying these two tasks. We observe that understanding and generation require fundamentally different granularities of visual information. This leads to a critical trade-off, particularly compromising performance in multimodal understanding tasks. TokenFlow addresses this challenge through an innovative dual-codebook architecture that decouples semantic and pixel-level feature learning while maintaining their alignment via a shared mapping mechanism. This design enables direct access to both high-level semantic representations crucial for understanding tasks and fine-grained visual features essential for generation through shared indices. Our extensive experiments demonstrate TokenFlow’s superiority across multiple dimensions. Leveraging TokenFlow, we demonstrate for the first time that discrete visual input can surpass LLaVA-1.5 13B in understanding performance, achieving a 7.2% average improvement. For image reconstruction, we achieve a strong FID score of 0.63 at 384×384 resolution. Moreover, TokenFlow establishes state-of-the-art performance in autoregressive image generation with a GenEval score of 0.55 at 256×256 resolution, achieving comparable results to SDXL. Our code and models are released at <https://github.com/ByteFlow-AI/TokenFlow>.

1. Introduction

Large Language Models (LLMs) have revolutionized natural language processing through their unified autoregressive framework, demonstrating remarkable capabilities across diverse tasks [1, 2]. However, in the multimodal domain of vision and language, a fundamental divide persists

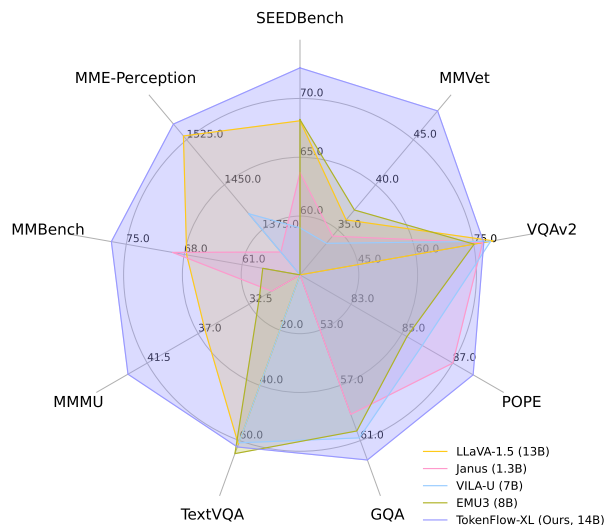


Figure 1. Multimodal Understanding Results with TokenFlow. We demonstrate for the first time that *discrete visual input* can surpass LLaVA-1.5 13B in understanding performance, achieving a 7.2% average improvement.

between perception and generation paradigms. Current approaches address them through distinct architectures: multimodal understanding models leverage vision encoders and projection layers to align visual representations with pre-trained LLMs [29, 52], while visual generation relies on either diffusion-based methods [39, 41] or discrete image tokens for autoregressive generation [38, 44, 51, 65]. This divergence motivates the pursuit of unified approaches capable of both understanding and generation.

The advent of GPT-4o [59] has greatly boosted interest in developing more generalist multimodal models. Early efforts to unify perception and generation capabilities [27, 46] have primarily focused on equipping LLMs with the power of diffusion models. However, these approaches introduce substantial architectural complexity and computational overhead, highlighting the need for a more elegant unified solution. Recent efforts have explored one

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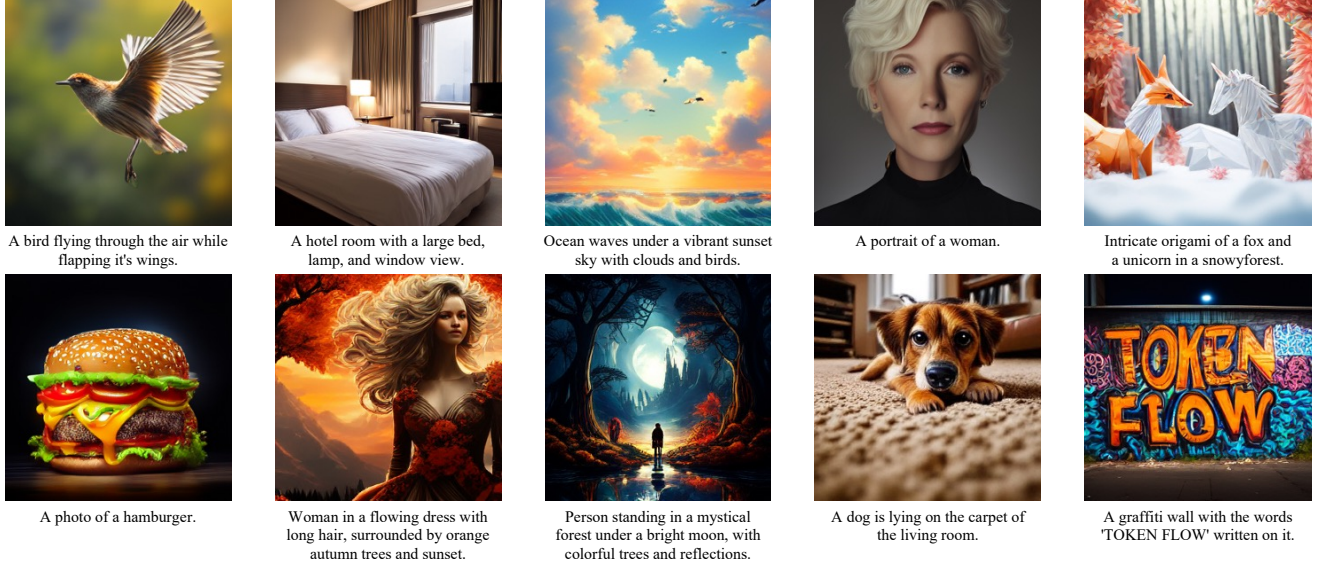


Figure 2. Visual Generation Results with TokenFlow. We present diverse 256x256 results across various styles, subjects, and scenarios.

promising direction: using a single transformer architecture to unify visual and textual information within the next-token prediction framework [48, 55]. This approach relies on VQ encoders to convert visual inputs into discrete tokens that can be processed alongside text, offering a potentially simpler and more efficient framework. By treating both modalities as sequences of discrete tokens, this framework enables end-to-end training within a single architecture.

However, a fundamental challenge exists in such unified approaches. Multimodal understanding demands *rich semantic representations* to support complex reasoning, while visual generation, on the other hand, requires *precise encoding of spatial structure and textural details*. Current methods predominantly employ reconstruction-targeted VQ encoders [13, 73], which are primarily optimized for reconstruction fidelity. While this optimization makes them well-suited for generation tasks, it potentially limits their ability to capture the high-level semantic features crucial for understanding tasks. While Janus [57] attempts to address this conflict by employing separate encoders for understanding and generation tasks, this increases model complexity without fundamentally resolving the underlying representation disparity. These limitations underscore a critical gap in the field: the absence of a unified visual encoding mechanism that can effectively serve both perception and generation objectives. This motivates our central research question: ***Can one single image tokenizer derive representations suitable for both multimodal understanding and generation?***

To address this challenge, we propose TokenFlow, a novel unified image tokenizer that bridges the gap between understanding and generation through a unique dual-flow design. The key insight is to decouple the learning of semantic and pixel-level features while maintaining their

alignment through a shared index mapping. By mapping patches with both semantic and pixel-level similarities to identical indices, the quantized features can be directly applied to both autoregressive visual generation and multimodal understanding. Unlike concurrent approach that constrains different feature levels within a single codebook [60], TokenFlow’s dual-codebook design enables specialized learning while maintaining cross-level correlations through shared indices. This innovation allows simultaneous access to both semantic and pixel-level representations without compromising either aspect. Specifically, TokenFlow adopts a dual-encoder architecture coupled with corresponding specialized codebooks. The semantic encoder, learned from a CLIP-style teacher, provides strong semantic priors, while the pixel encoder captures detailed visual information. The extracted features are then quantized by minimizing the weighted summation of semantic and pixel-level distances, creating a joint representation space.

Our framework exhibits remarkable scalability, maintaining exceptional codebook utilization (95%+) even with large-scale codebooks of over 130K entries - substantially advancing beyond prior approaches [13] in both capacity and efficiency. TokenFlow also achieves a strong FID score of 0.63 at 384x384 resolution. For text-to-image synthesis, we establish a new state-of-the-art GenEval score of 0.55 at 256x256 resolution in the autoregressive paradigm while requiring significantly fewer sampling steps compared to existing methods like EMU3 [55] and LlamaGen [44]. On multimodal understanding benchmarks, TokenFlow achieves new state-of-the-art performance with minimal training overhead, surpassing LLaVA-1.5 13B by 7.2% on average - for the first time discrete visual inputs can outperform this strong baseline. These results validate To-

kenFlow’s effectiveness as a unified visual tokenizer that bridges the long-standing gap between understanding and generation tasks.

2. Related Work

2.1. Tokenization for Visual Generation.

Vector quantized (VQ) image tokenizers have played a crucial role in recent advancements in autoregressive image generation [28, 34, 44, 51, 65]. [54] proposed the VQVAE, quantizing patch-level features using the nearest codebook entry, with the codebook learned with the encoder-decoder structure through reconstruction loss. VQVAE-2 [40] advanced this framework through exponential moving average updates and a hierarchical multi-scale approach. VQGAN [13] further enhanced the architecture by incorporating adversarial and perceptual losses, yielding more precise and detailed representations. Recent advances in VQ tokenizers have focused on three main directions: improving reconstruction fidelity and generation quality [21, 64, 73], enhancing codebook utilization [64, 70, 76], and exploring novel architectures such as the multi-scale VQVAE [25, 51] for next-scale prediction of images. While these methods effectively preserve local details after quantization, they often struggle to capture semantic-level information, limiting their effectiveness in autoregressive multi-modal image understanding tasks. Our proposed TokenFlow addresses this limitation by introducing dual codebooks with shared mapping, achieving state-of-the-art performance in both autoregressive generation and multimodal understanding.

2.2. Tokenization for Unified Multimodal Understanding and Generation

Recent efforts have emerged to bridge the gap between multimodal understanding and generation [23, 48, 55, 57, 60, 62]. Approaches like Chameleon [48], EMU3 [55] and Show-o [62] employ VQ tokenizers [13, 66, 73] to encode images for both tasks. However, these methods typically require multimodal training from scratch and often suffer performance degradation in visual perception tasks due to limited semantic representation in their tokenized features. SEED-LLaMA [23] introduced a novel VQ tokenizer incorporating high-level semantics for understanding and utilize SD [41] as generation decoder. Janus [57] attempted to address the modality gap by employing separate tokenizers for understanding [69] and generation [44], though this leads to increased model complexity without fundamentally resolving the underlying challenge. Concurrent work [60] proposed a unified vision tower aligning discrete visual features with text during pre-training. However, their approach constrains low-level and high-level representations within a single flow, limiting the upper bound of downstream performance. In contrast, our work posits that the key to uni-

fying understanding and generation lies in learning a universal mapping. By defining dual codebooks with shared mapping, TokenFlow enables flexible combinations of low and high-level features, resulting in superior performance across all downstream tasks.

3. Method

3.1. Motivation

Table 1. Comparison of various visual encoders on multimodal understanding [14, 23, 43] within the LLaVA-1.5 framework. VQKD is distilled from CLIP ViT-B/14. "Sem." refers to semantic encoders that learn semantic-level representations, while "Pix." indicates pixel-level tokenizers that focus on low-level visual features.

# Exp.	Visual Encoder	Type	MME-P $\uparrow$	SEEDB $\uparrow$	TQA $\uparrow$
<i>Continuous:</i>					
1	CLIP ViT-B/14 [37]	Sem.	1460.9	64.1	53.4
<i>Discrete:</i>					
2	VQGAN [13]	Pix.	756.1	38.2	46.8
3	VQGAN-LC [76]	Pix.	744.8	38.2	45.7
4	LFQ [66]	Pix.	889.5	41.1	46.4
5	VQKD [35]	Sem.	1252.4	57.8	48.2

Unifying multimodal understanding and generation into a cohesive next-token prediction paradigm requires a VQ tokenizer for extracting indices from input images. While traditional VQ tokenizers [13, 54, 66, 76] excel at pixel-level image reconstruction, our investigation reveals a significant limitation in their image understanding capabilities. We conducted experiments utilizing these tokenizers as feature extractors within the LLaVA-1.5 [29] framework. As shown in Exp. 2-4 of Tab. 1, the performance of these discrete tokenizers consistently lags behind that of the continuous tokenizer CLIP ViT-B/14 [37]. We posit that this performance gap stems from their pre-training objectives, which primarily optimize towards better low-level reconstruction quality. Consequently, the extracted features mainly encode low-level information, lacking the semantic-level understanding, which is crucial for complex visual reasoning.

Another straight forward solution for unified understanding and generation can be distill discrete tokens from pre-trained CLIP [8, 37, 45, 69], and then equip it with image reconstruction capability. As demonstrated in Exp. 5, VQKD, distilled from CLIP ViT-B/14, substantially reduces the performance gap compared to other discrete tokenizers. We further conducted an experiment to reconstruct the original image from quantized features extracted by VQKD. The reconstructed images exhibited significant blurring and a evident loss of high-frequency details, as shown in Fig. 9. We attribute this outcome to the nature of VQKD’s encoder, which maps semantically close patches into same codebook index. As visualized in Fig. 4 (a), it tends to map images with same semantical meaning to the same codebook index,

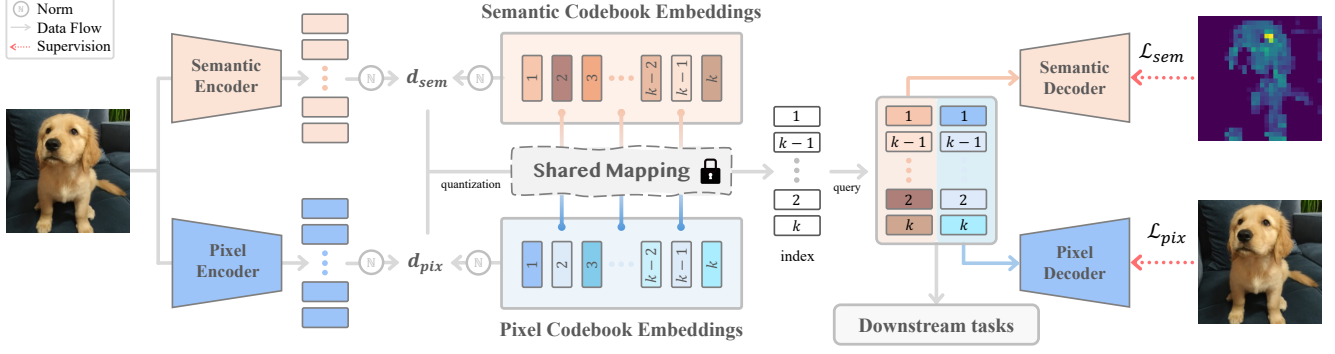


Figure 3. Overview of TokenFlow. We incorporate dual encoders and codebooks with a shared mapping, enabling the joint optimization of high-level semantics and low-level pixel details. For a given input image, distances d_{sem} and d_{pix} are calculated from the pixel-level and semantic-level codebooks, respectively, with the final codebook index and features determined by minimizing the weighted sum $d_{\text{sem}} + w_{\text{dis}} \cdot d_{\text{pix}}$. The resulting quantized features are independently decoded for both semantic alignment and image reconstruction training, and then concatenated to provide a unified representation for downstream tasks in understanding and generation.

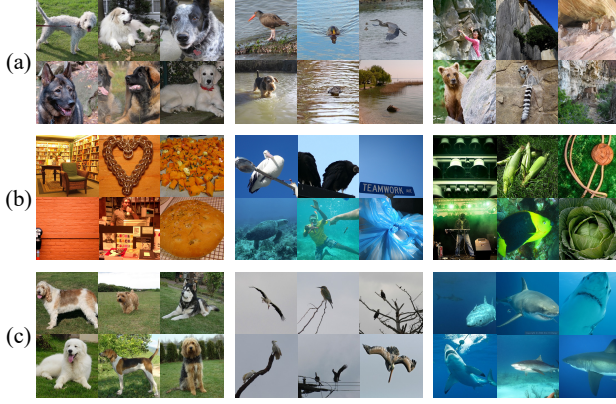


Figure 4. Visualization of images clustered by (a) VQKD [35], (b) VQGAN [13], and (c) Our TokenFlow. VQKD clusters exhibit semantic similarity, while VQGAN clusters exhibit low-level similarity (*i.e.* color). Our TokenFlow can successfully combine both semantic and low-level similarity. Implementation details of image clustering can be found in Appendix A.1.

while VQGAN (Fig. 4 (b)) tends to map visually similar images to the same codebook index, prioritizing low-level features over semantic content. Therefore, the reconstruction of fine-grained details from low-level dissimilar patches aggregated by VQKD becomes extremely challenging.

These observations highlight the necessity of developing a novel tokenization approach that can effectively handle high-level semantic understanding and low-level visual reconstruction tasks.

3.2. Unified Image Tokenizer

To bridge this gap, we propose TokenFlow (Fig. 3), a novel unified image tokenizer that enables joint representation learning at both semantic and pixel level. We find the key to unifying understanding and generation lies in learn-

ing an universal mapping. If the tokenizer can map patches that are both high-level and low-level similar to the same codebook index, then the quantized features can be easily decoded and directly applied to both autoregressive visual generation tasks and multimodal understanding tasks.

Encoder. Unlike previous approaches that utilize one single encoder to extract low-level image information, we propose a dual-encoder architecture comprising a semantic encoder $\mathcal{E}_{\text{sem}}$ and a pixel encoder $\mathcal{E}_{\text{pix}}$. This design enables the extraction of two distinct types of image features. For the semantic encoder, we initialize it with a pre-trained text-aligned vision encoder (e.g., CLIP ViT-B/14). This initialization strategy facilitates better learning of high-level text-aligned embeddings in the semantic codebook, ultimately enhancing the model’s multimodal understanding capabilities. For brevity here, we omit the spatial indices of feature representations, where $\hat{z}_{\text{sem}} = \mathcal{E}_{\text{sem}}(x) \in \mathbb{R}^{d_{\text{sem}}}$ and $\hat{z}_{\text{pix}} = \mathcal{E}_{\text{pix}}(x) \in \mathbb{R}^{d_{\text{pix}}}$ are the encoded features from semantic and pixel encoder.

Quantization. We introduce an innovative quantization approach that employs dual codebooks: semantic-level embeddings $\mathbf{Z}_{\text{sem}} = \{z_{\text{sem},i}\}_{i=1}^K \in \mathbb{R}^{K \times d_{\text{sem}}}$ and pixel-level embeddings $\mathbf{Z}_{\text{pix}} = \{z_{\text{pix},i}\}_{i=1}^K \in \mathbb{R}^{K \times d_{\text{pix}}}$, where K is the number of codebook entries. These two codebooks share a unified mapping, enabling simultaneous consideration of high-level semantic information and low-level pixel details during the quantization process. Given the encoded feature representations $\hat{z}_{\text{sem}}$ and $\hat{z}_{\text{pix}}$, we compute the distances to their respective codebook embeddings after l_2 -norm [64]:

$$d_{\text{sem},i} = \|\hat{z}_{\text{sem}} - z_{\text{sem},i}\|_2^2, \text{ for } i = 1, \dots, K \quad (1)$$

$$d_{\text{pix},i} = \|\hat{z}_{\text{pix}} - z_{\text{pix},i}\|_2^2, \text{ for } i = 1, \dots, K \quad (2)$$

$$i^* = \arg \min_i (d_{\text{sem},i} + w_{\text{dis}} \cdot d_{\text{pix},i}) \quad (3)$$

The optimal quantization index i^* is determined by minimizing the weighted sum of these two distances, where w_{dis} is the distance balance weight, as shown in Eq. (3). This joint optimization approach differs significantly from previous VQ methods that typically focus on learning the distribution of a single feature type. We further adopt the multi-scale VQ (MSVQ) structure [51] to enhance the richness of the codebook representation. Our shared mapping strategy enables the codebook to learn the joint distribution of high-level semantics and low-level features, resulting in several key advantages:

① **Scalability:** Our approach demonstrates consistent performance improvements in both generative and understanding tasks as the codebook size increases, since large codebook size offers more high- and low-level feature combination possibilities. With an expanded codebook size of 131,072, it can still maintain a remarkably high utilization rate of over 95% while achieving best image reconstruction quality and multimodal understanding performance.

② **Multi-task Capabilities:** By learning the joint distribution of semantic and pixel-level features, our method bridges the gap between generation and understanding tasks. This unified representation enables a single tokenizer to excel in both domains. This design also allows seamless integration of more codebooks to embed other type of feature representations, enabling extensibility to more downstream tasks without architectural modifications.

Decoder and Training Objective. Our architecture incorporates two distinct decoders, including semantic decoder $\mathcal{D}_{\text{sem}}$ and pixel decoder $\mathcal{D}_{\text{pix}}$ for reconstructing semantic features and original image. We employ a teacher model [35] (identical to the semantic encoder’s initialization) for target feature extraction. The semantic loss $\mathcal{L}_{\text{sem}}$ is computed as the l_2 distance between decoded and teacher-extracted features. The reconstruction loss is formulated as:

$$\mathcal{L}_{\text{pix}} = \ell_2(x, \hat{x}) + \mathcal{L}_{\text{p}}(x, \hat{x}) + \lambda_{\text{G}} \mathcal{L}_{\text{G}}(\hat{x}) \quad (4)$$

where $\hat{x} = \mathcal{D}_{\text{pix}}(z)$, ℓ_2 represents pixel-wise reconstruction loss, $\mathcal{L}_{\text{p}}(\cdot)$ denotes perceptual loss using LPIPS, and $\mathcal{L}_{\text{G}}(\cdot)$ represents adversarial loss with λ_{G} as its weight coefficient. Following vector quantization conventions, we employ a straight-through gradient estimator: $z = \text{sg}[z - \hat{z}] + \hat{z}$ where $\text{sg}[\cdot]$ denotes the stop-gradient operation. The codebook learning objective is: $\mathcal{L}_{\text{VQ}} = \|\text{sg}[\hat{z}] - z\|_2^2 + \beta \|\hat{z} - \text{sg}[z]\|_2^2$ where the second term represents commitment loss with balancing factor β . The total training objective is the sum of all losses: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sem}} + \mathcal{L}_{\text{VQ}} + \mathcal{L}_{\text{pix}}$.

3.3. Visual Generation with TokenFlow

TokenFlow helps us achieve SOTA performance in autoregressive text-to-image generation using the next-scale



Figure 5. Qualitative comparison of different sampling strategies in our framework. (a) Single-pass top- k ($k=1200$) and top- p ($p=0.8$) sampling exhibits inconsistent patterns and artifacts. (b) Our proposed multi-step sampling strategy produces more coherent and visually appealing results. Best zoomed in for details.

prediction paradigm. Below, we detail our training and inference strategy for high-quality image synthesis.

Training Strategy. Our visual generation architecture builds upon a pre-trained LLM model [53]. For text encoding, we leverage the model’s native BPE tokenizer to transform input text into discrete token sequences and extract feature representations. The original vocabulary is extended with specialized visual tokens. We extract the image tokens using TokenFlow, pass it through a MLP, and concatenate it with text tokens for training. Given the model’s autoregressive nature, we employ cross-entropy loss computed exclusively on image tokens. To enable classifier-free guidance [17] during inference, we randomly replace conditioned text with an empty string with probability $p_{\text{drop}} = 0.1$ during training. Following [11, 48, 56], we incorporate QK-normalization and norm re-ordering to enhance training stability and prevent loss spikes.

Inference Strategy. We observed that conventional top- k -top- p sampling strategies, when employed in the next-scale paradigm, often lead to image collapse and repetitive local patterns. This can be attributed to the cross-entropy training objective, which establishes attention-based relationships primarily with the top-1 prediction. Independent top- k sampling for each token during inference can result in tokens lacking direct correlations, leading to inconsistent or repetitive patterns that can only be partially remedied through subsequent scales’ attention. This issue becomes more severe particularly with limited inference steps.

To address this fundamental limitation, we propose a novel multi-step sampling approach: (i) Initial sampling: Perform top- k top- p sampling with parameters k_1 and p_1 . (ii) Refinement: Use the sampled output as input for a second round of sampling in the same scale with reduced parameters $k_2 < k_1$ and $p_2 < p_1$. This progressive narrowing of the sampling space maintains creative diversity while enforcing consistency through refinement steps. Empirical results demonstrate significantly more coherent and visually appealing generations compared to single-pass sampling methods (see Fig. 5 and detailed ablation in Appendix B.1).

3.4. Multimodal Understanding with TokenFlow

TokenFlow functions as a multi-scale VQ tokenizer, where the quantized multi-scale features can be directly fed into a pre-trained LLM for multimodal understanding training, following the LLaVA-1.5 [29] paradigm. The joint feature representations from dual flow serve as input to the model. We validate multiple feature input strategies: (i) Feature from all scales (ii) Final-scale feature only (iii) Residual features from all scales. We discover that features from the final scale achieves best overall performance, as detailed in Appendix B.1. This suggests that the final scale captures the most relevant semantic information for multimodal understanding, while additional scale features or residual features may introduce noise that compromises performance. Our model demonstrates substantial improvements over existing discrete multimodal methods. Notably, the performance gains can be achieved with minimal computational overhead, requiring less than 24 hour training on 8×A100 GPUs using LLaVA 1.5 training data.

4. Experiments

4.1. Experimental Setup

Datasets. TokenFlow is trained on LAION [42] and COYO-700M [5] and evaluate it on ImageNet [12]. To enhance face generation quality, we follow [48] and upsample the percentage of images with faces during tokenizer training by 2 times. For ablation studies, we train the tokenizer for 50 epochs on ImageNet-1K with CLIP ViT-B/14-224 [37]. For visual generation with TokenFlow, we trained it on a curated dataset of 60M high-quality images, with captions generated using Qwen-VL [3].

Implement Details. We employ three variants of TokenFlow (B/L/XL), using CLIP ViT-B/14-224 [37], ViTamin-XL-256 [8], and SigLIP-SO400M-patch14-384 [69] as respective teacher models and semantic encoder initializations. Detailed configurations are provided in Appendix A.2. For multimodal understanding, we employ Vicuna-v1.5-13B [10] and Qwen-2.5-14B [50] as the language backbone. For 256×256 visual generation training, we truncate captions to first sentence with 0.2 probability to enhance short prompt generation capabilities. The model is initialized with Llama-2-7b [53], and being trained for 2 epochs. At inference, we apply classifier-free guidance [17] with a scale factor of 7.5.

Evaluation Metrics. We assess reconstruction quality using rFID, PSNR, and SSIM on the ImageNet-1K validation set [12]. For multimodal understanding, we evaluate on a comprehensive suite of vision-language benchmarks: SEEDBench [22], MMVet [67], POPE [26], VQAv2 [16], GQA [19], TextVQA [43], AI2D [20], RealWorldQA [61], MMMU [68], MMBench [32], and MME [14]. Visual generation capabilities are evaluated using GenEval [15] and

DPG-Bench [18]. We opt not to include FID scores as argued that it does not correlate well with human assessment of the overall performance of generative models [7, 36, 46].

4.2. Unified Image Tokenizer

Table 2. Comparison of reconstruction quality on the ImageNet 50k validation set. “#Lvls.” represents the number of residual levels used. For 384×384 resolution, the downsample ratio of 14.2 is derived from 384/27.

Model	Res.	ratio	#Lvls.	rFID ↓	PSNR ↑	SSIM ↑
VQ-GAN [13]	256	16	1	4.98	20.00	0.629
LlamaGen [44]	256	16	1	2.19	20.79	0.675
RQ-VAE [21]	256	32	4	3.20	—	—
RQ-VAE [21]	256	16	4	1.30	—	—
VAR [51]	256	16	10	1.00	22.63	0.755
VILA-U [60]	256	16	4	1.80	—	—
Ours	256	16	9	1.37	21.41	0.687
LlamaGen [60]	384	14.2	1	0.94	21.94	0.726
VILA-U [60]	384	14.2	16	1.25	—	—
VAR [51]	384	16	13	2.09	22.73	0.774
Ours	384	14.2	15	0.63	22.77	0.731

In Tab. 2, we present reconstruction metrics of TokenFlow on 256×256 and 384×384 resolutions. The metric of VAR [51] is tested with the released checkpoint. At 256×256 resolution with a 16× compression ratio, TokenFlow achieves competitive performance with an rFID of 1.37, comparable to RQ-VAE while significantly outperforming previous methods such as VQ-GAN and LlamaGen. TokenFlow demonstrates superior reconstruction quality across all metrics in 384×384 resolution—a standard size in multimodal understanding tasks. These results validate the effectiveness of dual codebook design in preserving fine-grained visual details. Moreover, the incorporation of shared mapping enables TokenFlow to maintain high-level semantic features, as verified in Sec. 4.3.

4.3. Multimodal Understanding

TokenFlow, as a discrete visual encoder, demonstrates state-of-the-art performance across a comprehensive suite of multimodal understanding benchmarks. Following LLaVA-1.5’s training pipeline, we train TokenFlow-B and TokenFlow-L using LLaVA-Pretrain558K for adapter pre-training and LLaVA-v1.5-mix-665K for instruction tuning. For TokenFlow-XL, inspired by recent findings in [52], we leverage Cambrian-Alignment and Cambrian-10M for pretraining and instruction tuning respectively, as the teacher model SigLIP-SO400M benefits significantly from increased training data. As evidenced in Tab. 3, TokenFlow-XL achieves competitive or superior results compared to leading approaches with continuous inputs from CLIP-style encoders. Using the same language backbone (Vicuna 13B), TokenFlow-XL outperforms LLaVA-1.5 13B by

Table 3. Evaluation on multimodal understanding benchmarks. We collect evaluations including: SEEDB: SEED Bench-Img [22]; MMV: MM-Vet [67]; POPE [26]; VQAv2 [16]; GQA [19]; TQA: TextVQA [43]; AI2D [20]; RWQA: RealWorldQA [61]; MMMU [68]; MMB: MMBench [32]; MME [14] and MME-P: MME-Perception. We include approaches with continuous visual inputs (top) versus discrete visual inputs (bottom). The best results among approaches with discrete visual input are highlighted in bold. * results are not reported in original paper and tested with lmms-eval [71] using the released checkpoint. When calculating average, we use MME-P and divide it by 20 to have the same scale with other benchmarks.

Method	# Params	Res.	SEEDB	MMV	POPE	VQAv2	GQA	TQA	AI2D	RWQA	MMMU	MMB	MME	MME-P	Avg.
<i>Continuous Visual Input</i>															
InstructBLIP [30]	Vicuna-13B	224	58.8	25.6	78.9	—	49.5	50.7	—	—	—	36.0	—	1212.8	—
MiniGPT-4 [75]	Vicuna-13B	224	—	—	—	—	—	—	—	—	—	—	1158.7	866.6	—
BLIP-2 [24]	Vicuna-13B	224	46.4	22.4	—	—	—	42.5	—	—	26.6	—	—	1293.8	—
ShareGPT4V [9]	Vicuna-7B	336	69.7	37.6	—	80.6	63.3	60.4	58.0	54.9	37.2	68.8	1943.8	1567.4	—
NEXT-GPT [58]	Vicuna-7B	224	57.5	—	—	66.0	—	—	—	—	—	58.0	—	—	—
Qwen-VL-Chat [3]	Qwen-7B	448	57.7	—	—	78.2	57.5	—	—	—	—	—	1848.3	1487.5	—
Janus [57]	DeepSeek-LLM-1.3B	384	63.7	34.3	87.0	77.3	59.1	—	—	—	30.5	69.4	—	1338.0	—
LLaVA-1.5 [29]	Vicuna-13B	336	68.1	36.1	85.9	80.0	63.3	61.3	61.1	55.3	36.4	67.7	1826.7	1531.3	62.9
<i>Discrete Visual Input</i>															
Gemini-Nano-1 [49]	1.8B from scratch	—	—	—	—	62.7	—	—	—	—	26.3	—	—	—	—
Chameleon [48]	34B from scratch	256	—	—	—	69.6	—	—	—	—	—	—	—	—	—
LWM [31]	LLaMA-2-7B	256	—	9.6	75.2	55.8	44.8	18.8	—	—	—	—	—	—	—
SEED-LLaMA [23]	LLaMA-2-13B	224	53.7	—	—	63.4	—	—	—	—	—	—	—	—	—
Show-o [62]	Phi-1.5-1.3B	256	—	—	80.0	69.4	58.0	—	—	—	26.7	—	—	1097.2	—
VILA-U [60]	LLaMA-2-7B	256	56.3	27.7	83.9	75.3	58.3	48.3	—	—	—	—	—	1336.2	—
VILA-U [60]	LLaMA-2-7B	384	59.0	33.5	85.8	79.4	60.8	60.8	—	—	—	—	—	1401.8	—
EMU3 [55]	8B from scratch	512	68.2	37.2	85.2	75.1	60.3	64.7	70.0	57.4	31.6	58.5	1509.9*	1243.8*	60.9
TokenFlow-B	Vicuna-13B	224	60.4	22.4	84.0	70.2	59.3	49.8	54.2	49.4	34.2	55.3	1660.4	1353.6	55.2
TokenFlow-L	Vicuna-13B	256	62.6	27.7	85.0	73.9	60.3	54.1	56.6	49.2	34.4	60.3	1622.9	1365.4	57.5
TokenFlow-XL	Vicuna-13B	384	68.7	40.7	86.8	77.9	62.7	61.5	66.7	53.7	38.7	68.9	1840.9	1545.9	64.0
TokenFlow-XL	Qwen-2.5-14B	384	72.6	48.2	87.8	77.6	62.5	62.3	75.8	56.6	43.2	76.8	1922.2	1551.1	67.4

1.7% on average, for the first time demonstrates that model with discrete visual input can surpass this strong baseline. By simply changing the LLM backbone to Qwen-2.5-14B [50], we further surpass LLaVA-1.5 by 7.2%.

When compared to methods using discrete inputs, our approach demonstrates superior performance while maintaining training efficiency. Unlike models trained from scratch such as Chameleon and EMU3, our method requires less than 24 hour of training on 8×A100 GPUs using LLaVA 1.5 data. TokenFlow-XL 14B significantly outperforms EMU3 with an overall improvement of 10.7%. Given these promising empirical results, we position TokenFlow as a potential next-generation vision tokenizer for unified understanding and generation tasks. Our findings suggest that discrete visual representations can not only match but exceed the performance of continuous counterparts while maintaining practical training requirements.

4.4. Visual Generation

We evaluate our model’s generation capabilities against state-of-the-art methods including diffusion-based, autoregressive-based, and hybrid approaches on standard benchmarks GenEval [15] and DPG-Bench [18]. As shown in Tab. 4, our approach achieves competitive performance while requiring significantly fewer generation steps.

For 256×256 image generation, we employ a multi-step sampling strategy instead of the original 9-step sampling (one per tokenizer scale). Specifically, we apply three steps per scale with top- $k=[1200, 100, 1]$ and top-

$p=[0.8, 0.8, 1.0]$ across all scales except the first, totaling 25 steps. Under this inference scheme, our model achieves a GenEval score of 0.55, surpassing prominent diffusion models like Stable Diffusion v2.1 and PixArt-alpha. More significantly, it surpasses autoregressive methods such as Chameleon, LlamaGen, and EMU3, which require thousands of inference steps. With prompt rewriting, our model achieves 0.63, approaching DALL-E 3’s performance. On DPG-Bench, it achieves an average score of 72.9, outperforming LlamaGen, Show-o, SD v1.5, and PixArt-alpha. Moreover, our model only requires 2.7 seconds to infer one image with 1×A100 GPU, which is significantly faster than other autoregressive-based methods.

We further conduct additional text-to-image comparison between TokenFlow and the released VAR tokenizer [51]. Under identical training configurations and dataset settings, our model consistently demonstrates better performance across all benchmark metrics, this further showcasing the effectiveness of our unified tokenization approach.

4.5. Ablation Studies

Effect of Codebook Size. In Fig. 6, we experimented the impact of codebook size in our unified tokenizer, varying from 8,192 to 131,072. Our evaluation spans reconstruction quality, class-conditional generation, and multimodal understanding capabilities. For class-conditional generation, we employ the VAR transformer [51] with $d=16$, resulting in approximately 310M parameters. Notably, our approach maintains a consistently high codebook utilization rate ex-

Table 4. Comparison of generation quality on GenEval [15] and DPG-Bench [18]. "Step": the number of model runs needed to generate an image. † result is with rewriting.

Model	Text Pretrain	Res.	#Steps	GenEval Overall ↑	DPG-Bench Average ↑
<i>Diffusion-based</i>					
SD v1.5 [41]	CLIP ViT-L/14	512	50	0.43	63.18
DALL-E 2 [39]	CLIP ViT-H/16	1024	—	0.52	—
SD v2.1 [41]	CLIP ViT-H/14	768	50	0.50	—
SDXL [36]	CLIP ViT-bigG	1024	40	0.55	74.65
PixArt-alpha [7]	Flan-T5-XXL	512	20	0.48	71.11
DALL-E 3 [4]	Flan-T5-XXL	1024	—	0.67†	83.50
<i>Autoregressive meets diffusion</i>					
Show-o [62]	Phi-1.5	256	16	0.53	67.27
Transfusion [74]	—	256	250	0.63	—
<i>Autoregressive-based</i>					
Chameleon [48]	—	512	1024	0.39	—
LlamaGen [44]	Flan-T5-XL	512	1024	0.32	64.84
EMU3 [55]	—	512	4096	0.54 / 0.66†	80.60
VAR [51]	—	256	28	0.53	71.08
Ours	—	256	25	0.55 / 0.63†	73.38

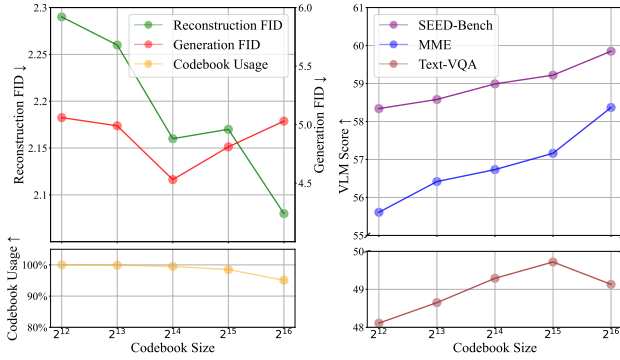


Figure 6. Impact of codebook size on reconstruction quality, class-conditional generation, and multimodal understanding benchmarks. MME is divide by 28 to have the same scale.

ceeding 95% even with codebook size of 131,072, attributed to our shared mapping design. The shared mapping allows for effective combinations of high-level semantic features and low-level details, addressing a common limitation of conventional VQ tokenizers [13] that typically suffer from deteriorating utilization rates at larger scales.

Our results reveal that increasing codebook size enhances performance across multimodal understanding benchmarks and reconstruction quality. However, when codebook size exceeds 32,768, we observe a slight degradation in class-conditional generation performance. This phenomenon can be attributed to the increased complexity of learning for autoregressive generation with larger codebooks. Based on this finding, we adopt a codebook size of 32,768 for our text-to-image generation experiments.

Effect of Key Design Choice. We validate the effectiveness of our key design choices in TokenFlow: shared mapping, multi-scale vector quantization (MSVQ), and CLIP initialization for the semantic encoder. As shown in Tab. 5,

we start with a baseline that uses one single codebook distilled from CLIP ViT-B/14, coupled with a pixel decoder for direct image reconstruction from semantic features. This baseline yields a high reconstruction FID of 8.07, primarily due to the challenge of reconstructing fine-grained pixel details solely from semantic features, as visualized in Fig. 9. The introduction of shared mapping (Row 2) enables the two codebooks to capture high-level and low-level features simultaneously. By weighted distance computation, we quantize the input with optimal combinations of high-level and low-level features. This design significantly improves reconstruction quality (-4.11 rFID) while maintaining comparable understanding capabilities.

We further find that incorporating MSVQ [51] (Row 3) introduces multi-granular information into the codebook embeddings, which results in enhanced reconstruction performance, with rFID of 2.18. Moreover, this hierarchical design enables a next-scale prediction paradigm in downstream text-to-image generation tasks, offering significant inference speed advantages over traditional next-token prediction approaches [47, 51]. Initializing the semantic encoder with pretrained CLIP weights (Row 4) while making it unfrozen during tokenizer training provides strong semantic priors for codebook embeddings. This results in substantial improvements across all understanding metrics. Given these empirical results, we adopt this configuration as our final model architecture and extend our experiments with stronger teacher models, additional training data, and longer training iterations.

Table 5. Impact of key design choices on reconstruction quality and multimodal understanding benchmarks. Best results for each metric are highlighted in bold.

Shared Mapping	MSVQ	CLIP Init.	rFID ↓	MME-P ↑	SEEDB ↑	TQA ↑
✓			8.07	1252.38	57.84	49.16
✓			3.96	1212.51	55.97	47.42
✓	✓		2.18	1209.90	56.08	47.40
✓	✓	✓	2.16	1312.09	58.99	49.29

5. Conclusion

In this work, we introduce TokenFlow, a novel unified image tokenizer that effectively bridges the gap between multimodal understanding and generation through its innovative dual-codebook architecture. By decoupling semantic and pixel-level feature learning while maintaining their alignment via shared mapping, TokenFlow successfully addresses the fundamental issue between different granularities of visual information required for understanding and generation tasks. Experiments demonstrate state-of-the-art reconstruction quality, efficient multimodal understanding with minimal training costs, and fast high-quality generation, validating its unified approach and positioning TokenFlow as a promising next-era foundation tokenizer for vision-language systems.

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